



Precision IoT

Complete guide to monitoring your sensor network, managing devices, and reviewing intelligent ML-generated alerts from the mobile app.

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Getting Started

Follow these steps the first time you open the app, whether you are signing in with Google or creating a email/password account.

Signing In

1 Open the App

Launch Precision IoT on your mobile device. The login screen appears with two sign-in options.

2 **Sign In with Google (recommended)**

Tap **Sign in with Google**. Choose your Google account from the picker. Your name and photo load automatically.

3 **— or — Sign In with Email & Password**

Tap **Sign in with Email**, enter your registered email and password, then tap **Sign In**. This path is for accounts that have already been created and approved by an administrator.

4 **Automatic Redirect to Dashboard**

On success you are taken straight to the Dashboard. Your profile is synced in the background.

Creating a New Email/Password Account

1 **Open the Sign-Up Popup**

On the login screen, tap **Sign in with Email**, then choose the **Sign Up** option in the popup.

2 **Enter Required Details**

Provide a **email address**, your display name, and a password with at least 6 characters.

3 **Submit and Wait for Confirmation**

Tap **Submit**. The app creates your account and sends the new credentials to the backend. A success message appears only after the backend acknowledges receipt.

4 **Read the Success Message**

When registration completes, the app shows **New user created successfully, wait for admin approval**. At this stage the account exists, but app access is still blocked.

5 **Wait for Administrator Approval**

An administrator must unblock your account in the admin portal before your first email/password login will succeed.

6 **Sign In After Approval**

Once approved, return to **Sign in with Email** and log in with the same email and password to reach the Dashboard.

New user?

Your first Google Sign-In automatically creates an active account. If you want an email/password account, sign-up flow above.

Pending approval or blocked account

If your account is waiting for administrator approval, or an administrator has blocked it later, the Dashboard will remain inaccessible until the account is unblocked.

Maximum 3 mobiles per account

Each account can be signed in on at most

3 different phones

. If you try to sign in on a 4th device, the app shows a "Device limit reached" message and signs you out. Sign out from one of your other phones first, then try again.



Dashboard Overview

The Dashboard is the central hub of the app. Two tabs sit at the top; a profile icon is in the header.



Devices Tab

Lists all your connected Raspberry Pi sensor devices. Add new devices by scanning their QR code, view live status, and manage sensors or cameras.



Alerts Tab

Displays the real-time ML Intelligence Feed. Review, rate, and manage automated detection alerts.



Network Health

At the bottom of the Devices tab — global latency meter and live bar chart of connectivity across your device fleet.



Push Notifications

Critical alerts are pushed to your device instantly, even when the app is in the background.



Managing Devices

Adding a Device (Scan to Add)

Devices are added by scanning the QR code shown by the Raspberry Pi. This is the only way to add a device — there is no browsable list. Scanning a valid QR proves you have physical access to that specific device.

1 Display the QR Code on the Device

On the Raspberry Pi, run its device registration. The Pi generates and shows a pairing QR code on its attached screen.

2 Tap “Scan to Add”

On the Devices tab, tap the **Scan to Add** button. The app opens your phone camera.

3 Point the Camera at the QR Code

Line up the QR code within the on-screen frame. The app scans it automatically — no shutter button needed.

4 Device Added

On success the device is paired to your account and appears in your list immediately.

 Camera permission

The first time you tap Scan to Add, allow camera access when prompted so the scanner can read the QR code.

 Maximum 3 devices per account

Each account can have at most **3 devices** paired at once. To add a new device beyond that, remove one first using the REMOVE action.

 QR codes expire

A pairing QR code is valid for a limited time. If scanning fails with an “expired” or “invalid” message, re-run the registration on the Pi to display a fresh QR code and scan again.

Reading a Device Card

Field	What it means
Device Name	Human-friendly label you assigned (e.g., “Front Gate Camera”)
Device ID	Unique system identifier (auto-generated)
Status	ACTIVE = online OFFLINE = unreachable
Last Pulse	Time since the device last contacted the server (e.g., “2 min ago”)
Uptime	Availability percentage since device was added (e.g., “99.7%”)

Device Actions

-  **SENSORS** Open the sensor management panel for this device.
-  **CAMERA** Open the live WebRTC video stream for this device in your phone browser.
-  **RENAME** Change the display label for easier identification.

**REMOVE**

Un-claim this device from your account. It can be re-claimed later; no hardware is deleted.



Sensor Control

Tap **SENSORS** on any device card to open that device's sensor management panel.

Available Operations

Action	How To	Effect
View sensors	Open the SENSORS panel	Lists every sensor with its label, GPIO pin, and ON/OFF state
Toggle ON / OFF	Tap the ON or OFF button next to a sensor	Sends a remote command to enable or disable that sensor on the device



Access required

You can only control sensors on devices you own or have been granted controller-level access to by an administrator.



Live Camera Streaming

Each Raspberry Pi device can stream live video using WebRTC. Tapping CAMERA opens the live stream in your phone's web browser.

Starting a Live Stream

1 Tap CAMERA on a Device Card

A confirmation dialog appears. Tap **Open Stream** to continue.

2 Browser Opens the Stream

Your phone's browser opens the live camera viewer page for that device and begins connecting.

3 Stream Connects and Plays

The status shows "Connecting to camera..." then "Connected to camera feed" once the live video appears.

4 Return to the App

When finished, switch back to the app from your browser to continue.

What the Viewer Shows

Item	Details
Connecting status	"  Connecting to camera..." while it establishes the link
Connected status	"  Connected to camera feed" once live video is playing
Timeout message	"  Connection timeout" if the device's camera is not currently online

Device camera must be running

The stream only connects if the Raspberry Pi is powered on, online, and its camera service is running. If you get a connection timeout, verify the device is active and the camera is enabled.



ML Alerts & Ratings

The **Alerts tab** shows a live feed of events detected by the machine-learning model running on your sensor devices.

Alert Retention Settings

Choose how long alerts are kept before automatic deletion. Tap your preferred period at the top of the Alerts tab:

7 days 15 days 30 days **Never**

Your selection saves automatically. “30 days” is the default. Use *Never* only for critical records.

Reading an Alert Card

Component	Description
Risk Badge	CRITICAL HIGH MEDIUM LOW
Device Name	Which sensor device triggered the alert
Detected Objects	What the ML model found (e.g., “person”, “vehicle”, “fire”)
Preview Image	Thumbnail captured at the moment of detection (tap to enlarge)
Timestamp	Relative time since alert (e.g., “3 min ago”)
Accuracy Status	✓ Accurate or ✗ Inaccurate once rated
Rating	★ your 1–10 score (shown after rating)

Risk Level Colour Guide

-  **CRITICAL** — Immediate action required
-  **HIGH** — Prompt attention needed
-  **MEDIUM** — Review when possible
-  **LOW** — Informational only

Rating an Alert (Feedback)

- 1

Tap an Alert Card
Opens the Rating dialog for that alert.
- 2

Mark Accuracy
Tap **Accurate** if the detection was correct, or **Inaccurate** for a false positive.
- 3

Select a Score (1–10)

Tap a number to rate the usefulness of the detection. 10 = very useful.

4 Tap OK / Submit

Saved instantly. Card background updates to green (accurate) or red (inaccurate).

Why rate alerts?

Every rating is used to retrain the ML pipeline. Consistent feedback leads to fewer false alarms within days.

Viewing Full Alert Details

Tap **VIEW DETAILS** on any alert card to open a modal with:

- Full-resolution capture image from the moment of detection
- Complete detection metadata: detected classes, confidence, bounding boxes
- Original risk level assigned by the model



Push Notifications

The app pushes a notification to your device the moment a new ML alert is generated, even when the app is not open.

Enabling Notifications

1 Allow on First Launch

The app requests notification permission on sign-in. Tap **Allow**.

2 Re-enable if Denied

Go to phone *Settings* → *Apps* → *Precision IoT* → *Notifications* and turn them on.

Notification Format

Part	Example
Title	 CRITICAL
Body	person, vehicle — raspberrypi-front-gate
Sound & Vibration	Enabled by default for all risk levels
Tap action	Opens the app on the Alerts tab

 No duplicates

Each alert triggers exactly one notification, even if the app is reopened multiple times.



Profile & Sign Out

Viewing Your Profile

Tap the profile icon in the top-right of the Dashboard to open your profile panel. It shows your display name, profile photo, registered email, and sign-in method.

Signing Out

1 Open Profile Panel

Tap the profile icon in the top-right corner.

2 Tap Sign Out

Confirm in the dialog. Session is cleared and the app returns to the Login screen.

 Your data is safe

Signing out does not delete devices or alert history. Everything is restored on next sign-in.



Blocked Account Access

If your account is pending administrator approval or has been blocked, the app prevents Dashboard access until an administrator restores access.

What you see	What it means
Account pending activation	Your email/password account was created successfully and is still waiting for administrator approval
Access Denied screen	A system administrator has blocked an account that was previously active
Block reason text	The reason provided by the administrator, if one was supplied
Sign Out button	Only available action — clears your local session

 Cannot access app while approval is pending or while blocked

You cannot view devices, sensors, or alerts until an administrator approves or restores your access. Contact your system administrator if approval is delayed.



Troubleshooting

Sign-in fails or loops

- Ensure a stable internet connection.
- Use the correct Google account (the one with access).
- Close and reopen the app, then try again.
- For email/password: check caps lock and trailing spaces.

“Device limit reached” on sign-in

- Your account is already signed in on the maximum of 3 phones.

- Sign out from one of your other phones, then sign in on this one.

Cannot add a device / “Device limit reached” when scanning

- You already have 3 devices paired. Remove one with the REMOVE action before adding another.
- If the QR code is rejected as expired or invalid, re-run the device registration on the Pi to show a fresh QR code.
- Make sure you allowed camera permission for the scanner.

Device shows OFFLINE

- Confirm the Raspberry Pi is powered on.
- Verify Wi-Fi / internet connectivity on the device.
- Restart the device and wait 30 seconds to reconnect.
- If persistent, contact your system administrator.

Camera stream won't load / connection timeout

- Confirm the device status is **ACTIVE** and its camera service is running.
- Check your own internet connection.
- Close the browser tab and tap CAMERA again to retry.
- If the browser did not open, allow the app to open external links.

Sensor toggle has no effect

- Confirm the device is online.
- Wait one poll cycle (~15 seconds) for state to propagate.
- Restart the device if state remains stuck.
- Make sure you have controller-level access to this device.

Push notifications not arriving

- Go to phone *Settings* → *Notifications* → *Precision IoT* and allow notifications.

- Sign out and back in to re-register your push token.
- Disable Battery Saver mode that may block background activity.

Alerts feed is empty

- Check your **Alert Retention** setting — old alerts may have been auto-deleted.
- Verify at least one device is online and the ML service is running.
- Pull down to refresh the Alerts tab.

Need Help?

If you encounter an issue not covered in this guide, contact your system administrator or the Precision IoT support team.

For technical documentation, API references, and admin configuration guides, refer to the full system documentation index.

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